

5 RC and LC Power Supply Filters

RC π Filter

→ The ripple voltage that appears across the reservoir capacitor in a rectifier power supply can be attenuated by the use of an additional resistor and capacitor which together function as an ac voltage divider.

C_1 = Reservoir Capacitor

R_1 and C_2 = Additional components

Combination of C_1 , R_1 and C_2 is referred to as a π filter, because of the π -shaped arrangement of the circuit components.

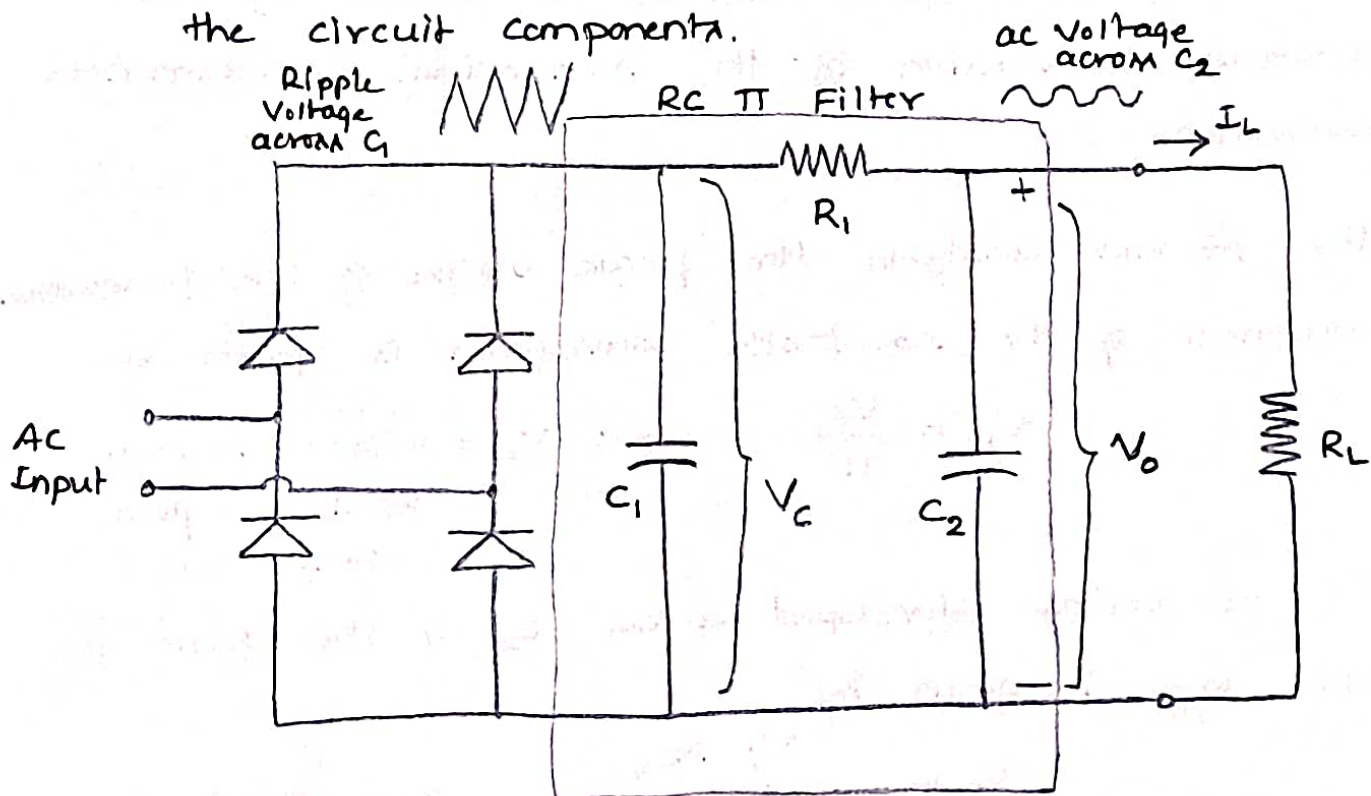


Fig. (a) RC π Filter circuit

→ Assuming a constant output load current, the reservoir capacitor continues to charge and discharge producing a sawtooth (ripple) waveform across C_1 , regardless of

The presence of the additional components

- The sawtooth waveform is composed of a fundamental ac voltage (same freq. as the ripple) and a number of smaller amplitude, high freq. harmonic components.
- Due to their higher frequencies, the harmonic components are severely attenuated than the fundamental freq. component by the voltage division ~~proportional~~ across R_1 and C_2 .
- This combined with the smaller input amplitude of the harmonics means that the waveform developed across C_2 (the filter output) is essentially an attenuated version of the sinusoidal fundamental component.
- By Fourier analysis, the peak value of the fundamental component of the sawtooth waveform is given by

$$V_p = \frac{V_r}{\pi} \quad \text{where } V_r = \text{ripple voltage peak-to-peak amplitude}$$

- The ac voltage developed across C_2 is the filter ac output and is given by

$$V_o = \frac{V_i X_{C_2}}{\sqrt{R_1^2 + X_{C_2}^2}}$$

where V_i is the filter ac input voltage applied across capacitor, C_1

$$X_{C_2} = \frac{R_1}{\sqrt{\left(\frac{V_i}{V_o}\right)^2 - 1}}$$

$$\text{if } \left(\frac{V_i}{V_o}\right)^2 \gg 1 \Rightarrow X_{C_2} = \frac{R_1}{(V_i/V_o)}$$

The additional attenuation of the ripple voltage depends upon the selection of R_1 and C_2 values

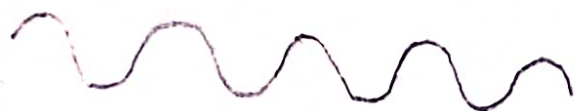
→ However, the dc voltage drop across R_1 produced by the output current (V_{R_1}) must be considered when determining a suitable resistance for R_1



Sawtooth waveform filter input voltage



Fundamental freq. Component

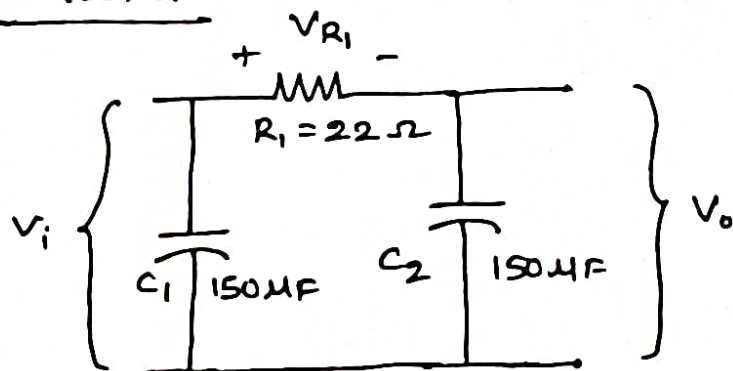


High-freq smaller amplitude components

Example 3.12

The 2V ripple waveform across capacitor C_1 in Ex. 3.8 is to be further attenuated by the use of additional resistor and capacitor. If $R_1 = 22\Omega$, $C_1 = C_2 = 150\mu F$, calculate dc output voltage and output ripple amplitude

Solution



$$V_o(\text{dc}) = E_o(\text{ave}) - I_L R_1$$

$$= 20\text{V} - 40\text{mA} \times 22\Omega$$

$$= 19.12\text{V}$$

$$V_i = V_p = \frac{V_r}{\pi} = \frac{2}{\pi} = 637\text{mV}$$

$$X_{C2} = \frac{1}{2\pi f_r C_2} = \frac{1}{2\pi \times 120 \text{ Hz} \times 150 \mu\text{F}}$$

$$X_{C2} = 8.84 \Omega$$

$$V_o = \frac{V_i X_{C2}}{\sqrt{R_1^2 + X_{C2}^2}} = \frac{637 \text{ mV} \times 8.84 \Omega}{\sqrt{(22 \Omega)^2 + (8.84)^2}}$$

$$V_o = 238 \text{ mV (peak)}$$

$$= 475 \text{ mV (peak-to-peak)}$$